

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

08

PRODUCTION RULES OF FSA

DENNIS GUNAWAN

REVIEW

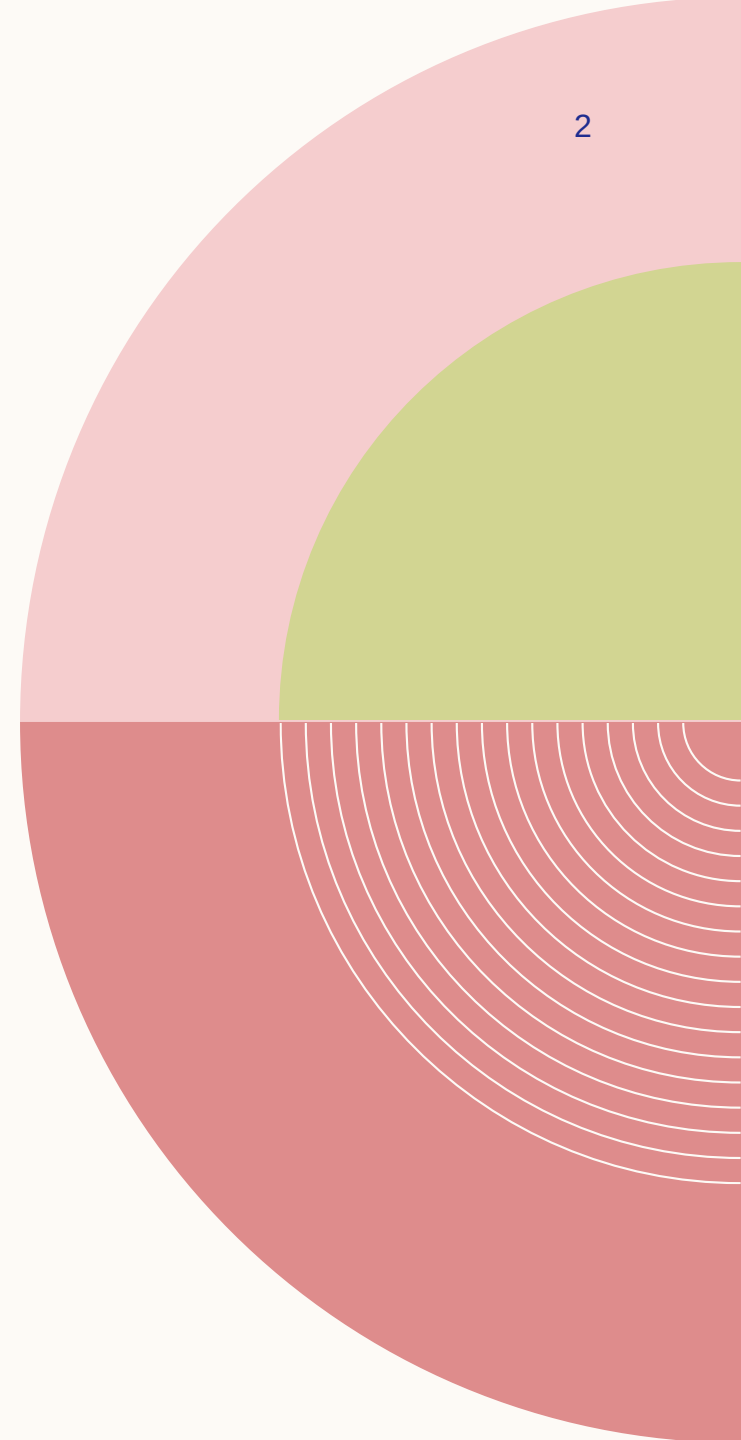
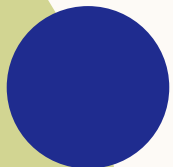
Regular Expression:

Identities of Regular Expression

Arden's Theorem

Conversion of FSA to Regular Expression

Conversion of Regular Expression to FSA



OUTLINE

Regular Grammar

Conversion of FSA to Regular Grammar

Conversion of Regular Grammar to FSA

CHOMSKY HIERARCHY

Grammar Type	Grammar Accepted	Language Accepted	Automata
Type 0	Unrestricted Grammar	Recursively Enumerable Language	Turing Machine
Type 1	Context Sensitive Grammar	Context Sensitive Language	Linear Bounded Automata
Type 2	Context Free Grammar	Context Free Language	Pushdown Automata
Type 3	Regular Grammar	Regular Language	Finite State Automata

GRAMMAR

- A grammar G can be formally described using 4 tuples as $G = (V, T, P, S)$
 - V : set of variables or non-terminal symbols
 - T : set of terminal symbols
 - P : production rules for terminals and non-terminals
 - S : start symbol
- A production rule has the form $\alpha \rightarrow \beta$ where α and β are strings on $V \cup T$ and at least one symbol of α belongs to V

REGULAR GRAMMAR

- Regular Grammar can be divided into two types

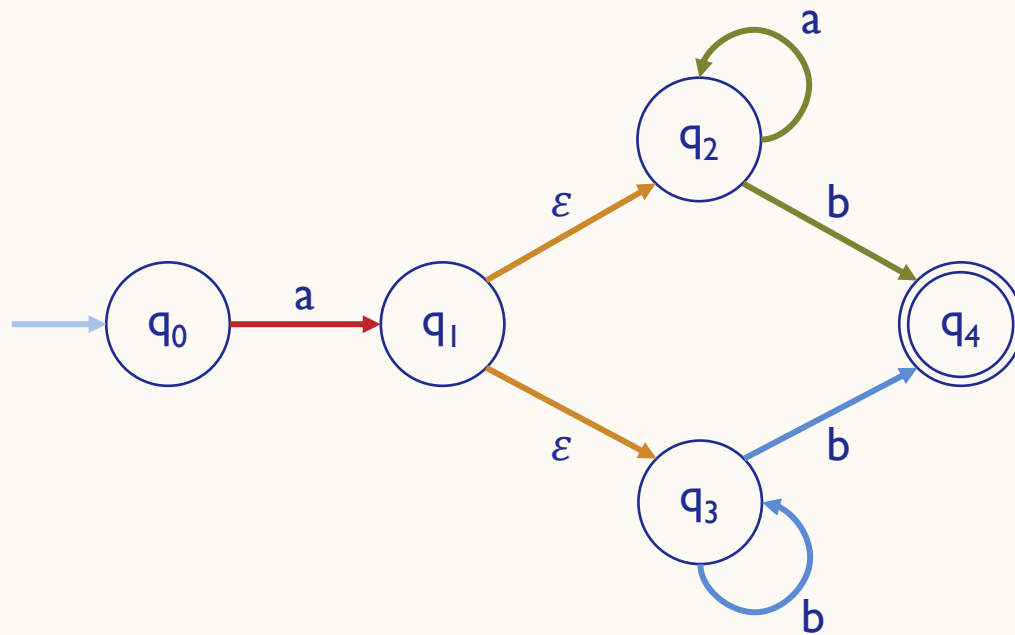
Right Linear Grammar	Left Linear Grammar
A grammar is said to be right linear if all productions are of the form $A \rightarrow xB$ $A \rightarrow x$ where $A, B \in V$ and $x \in T$ $S \rightarrow abS \mid b$	A grammar is said to be left linear if all productions are of the form $A \rightarrow Bx$ $A \rightarrow x$ where $A, B \in V$ and $x \in T$ $S \rightarrow Sab \mid b$

REGULAR GRAMMAR?

- $G = (\{S, S_1, S_2\}, \{a, b\}, P, S)$
 - $S \rightarrow S_1ab$
 $S_1 \rightarrow S_1ab \mid S_2$
 $S_2 \rightarrow a$
- $G = (\{S, A, B\}, \{a, b\}, P, S)$
 - $S \rightarrow A$
 $A \rightarrow aB$
 $B \rightarrow Ab$

CONVERSION OF FSA TO REGULAR GRAMMAR

Construct the regular grammar corresponding to the following FSA.



Production Rules

$S \rightarrow aA$

$A \rightarrow B \mid C$

$B \rightarrow aB \mid b$

$C \rightarrow bC \mid b$

Grammar

$V = \{S, A, B, C\}$

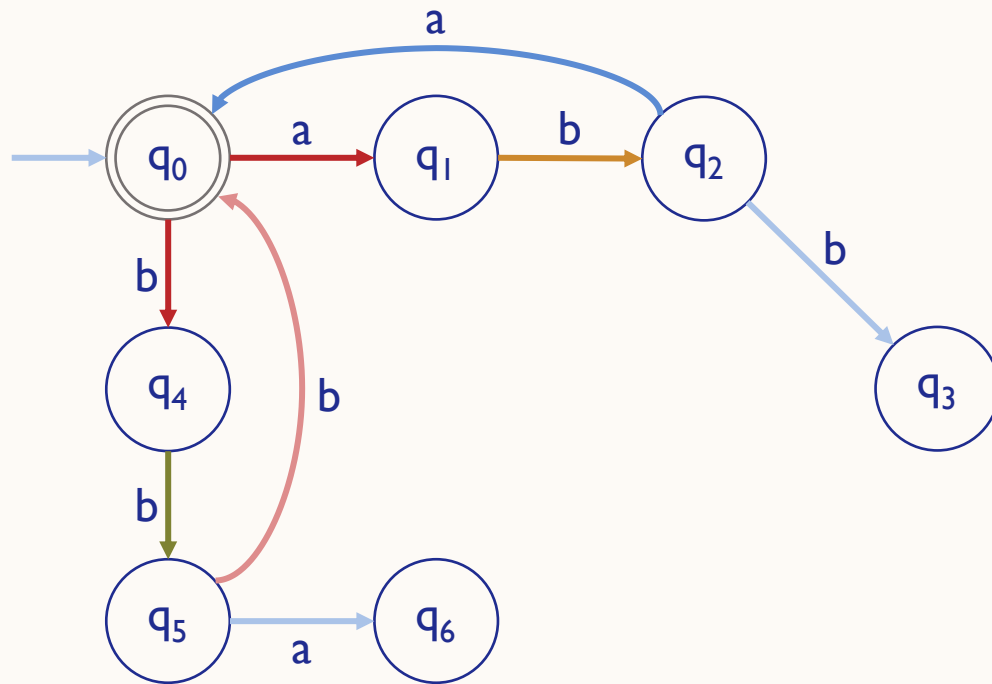
$T = \{a, b\}$

$P = \{S \rightarrow aA, A \rightarrow B \mid C, B \rightarrow aB \mid b, C \rightarrow bC \mid b\}$

$S = S$

CONVERSION OF FSA TO REGULAR GRAMMAR

Construct the production rules corresponding to the following FSA.



Production Rules

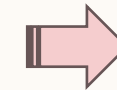
$$S \rightarrow aA \mid bB \mid \varepsilon$$

$$A \rightarrow bC$$

$$B \rightarrow bD$$

$$C \rightarrow aS$$

$$D \rightarrow bS$$



$$S \rightarrow aA \mid bB \mid \varepsilon$$

$$A \rightarrow baS$$

$$B \rightarrow bbS$$

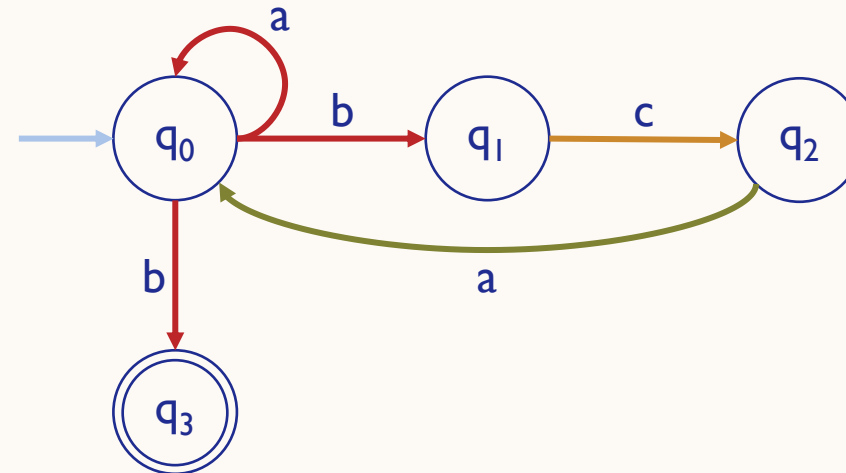
CONVERSION OF REGULAR GRAMMAR TO FSA

Construct the FSA corresponding to the following production rules of a regular grammar.

$S \rightarrow aS \mid bB \mid b$

$B \rightarrow cC$

$C \rightarrow aS$



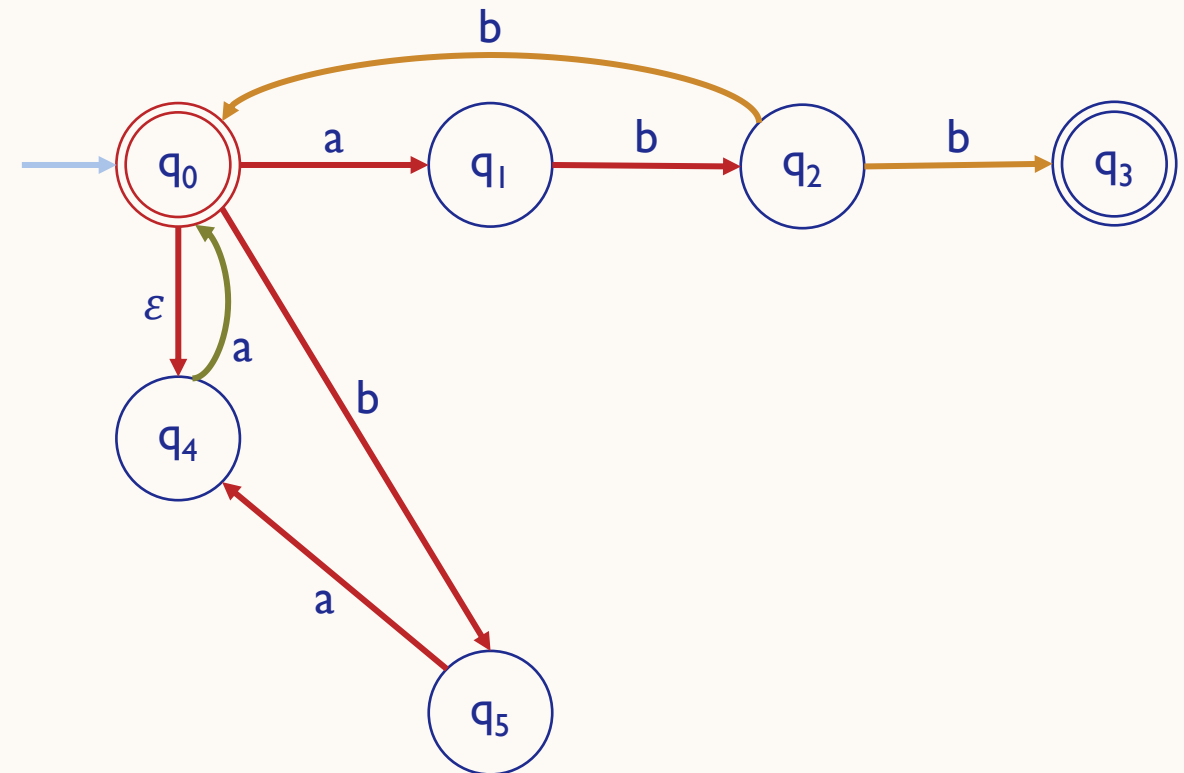
CONVERSION OF REGULAR GRAMMAR TO FSA

Construct the FSA corresponding to the following production rules of a regular grammar.

$S \rightarrow abA \mid B \mid baB \mid \varepsilon$

$A \rightarrow bS \mid b$

$B \rightarrow aS$

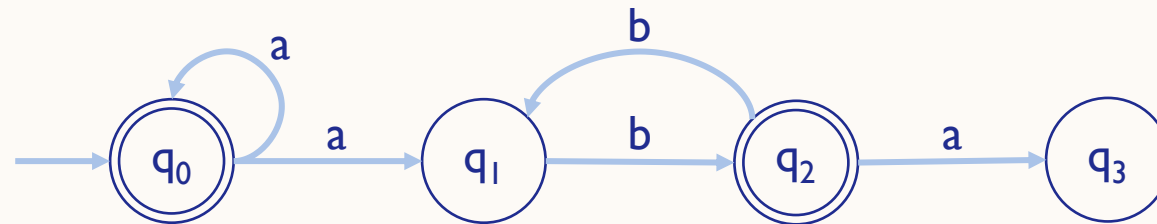




PRACTICE

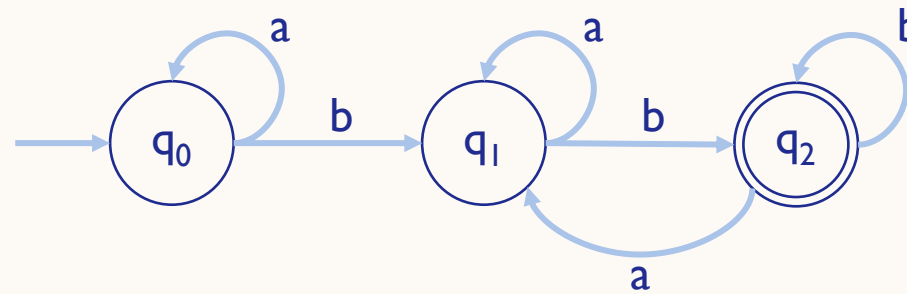
EXERCISES

1. Construct the regular grammar corresponding to the following FSA.



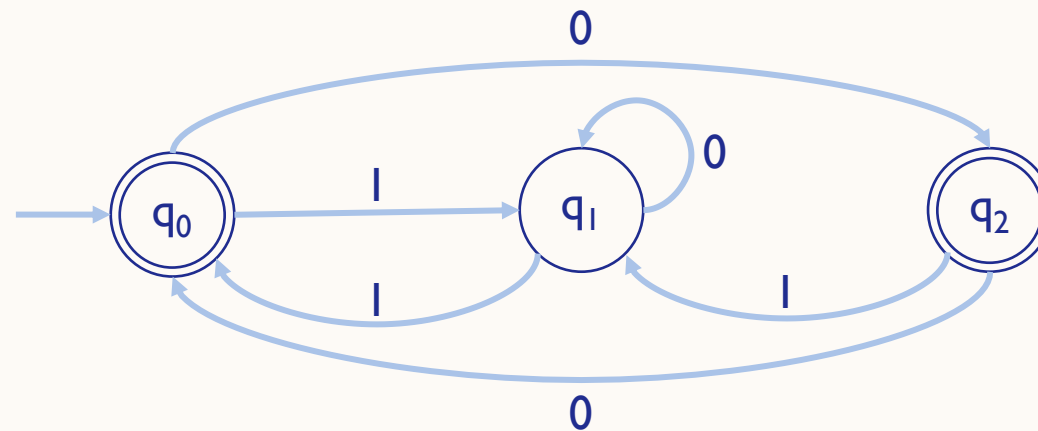
EXERCISES

2. Construct the regular grammar corresponding to the following FSA.



EXERCISES

3. Construct the regular grammar corresponding to the following FSA.



EXERCISES

4. Construct the FSA corresponding to the following production rules of a regular grammar.

$$S \rightarrow 0A$$

$$A \rightarrow 10A \mid \varepsilon$$

EXERCISES

5. Construct the FSA corresponding to the following production rules of a regular grammar.

$$A \rightarrow 0B \mid 1D \mid 0$$

$$B \rightarrow 0D \mid 1C$$

$$C \rightarrow 0B \mid 1D \mid 0$$

$$D \rightarrow 0D \mid 0$$

EXERCISES

6. Construct the FSA corresponding to the following production rules of a regular grammar.

$$S \rightarrow aB \mid bA \mid \varepsilon$$

$$A \rightarrow abaS$$

$$B \rightarrow babS$$

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

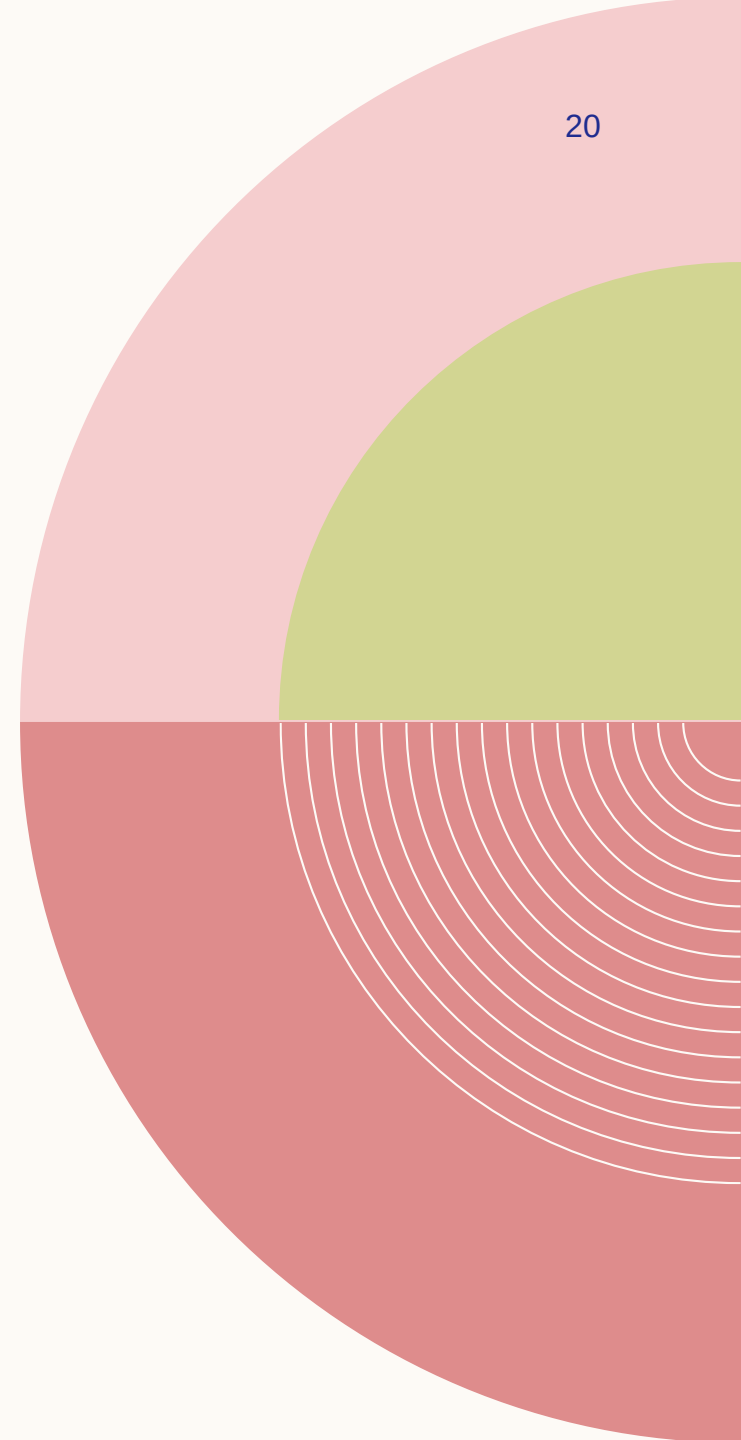
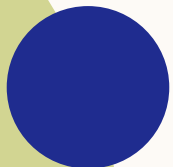
Derivation Tree:

Context Free Grammar

Derivations from a Grammar

Parsing

Ambiguous Grammar



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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